MIL-P-2067E
30 August 1988
SUPERSEDING
MIL-P-2067D
30 September 1968

MILITARY SPECIFICATION

PAILS, METAL, MEASURING (FOR BAKERIES)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers pails for measuring liquids and flour.

1.2 Classification. The measuring pails shall be of the following two types as specified (see 6.2).

Type I - Liquid measuring, 4-gallon capacity, graduated in gallons

Type II - Flour measuring, 40-pound capacity

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7240

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:17Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-A-250/2 - Aluminum Alloy 3003, Plate and Sheet
QQ-S-698 - Steel Sheet and Strip, Low Carbon
QQ-W-461 - Wire, Steel, Carbon (Round, Bare, and Coated)
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall

MILITARY

MIL-T-10727 - Tin Plating; Electrodeposited or Hot-Dipped, For Ferrous and Nonferrous Materials
MIL-C-52950 - Crate, Wood, Open and Covered

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MS 51931 - Rivet, Solid-Tinner's

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-11-272 - Pails, Metal, Measuring for Bakeries

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 20260-16T14:17Z
Check the source to verify that this is the current version before use.

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Material. Materials and components shall be as specified herein and on Drawing 5-11-272. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document.

3.2.1 Steel, sheet and strip. Steel sheet shall conform to cold rolled, commercial quality and steel strip shall conform to cold rolled, half-hard temper of QQ-S-698.

3.2.2 Wire steel. Steel wire shall conform to steel number 1010, 1015, or 1020, finish 3 of QQ-W-461.

3.2.3 Aluminum, sheet and strip. Aluminum alloy sheet and strip shall conform to 3003 of QQ-A-250/2.

3.2.4 Rivets. Tinner's rivets shall conform to MS 51931.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:17Z
Check the source to verify that this is the current version before use.

3.3 Design and construction. The pails shall be designed and constructed as specified herein and on Drawing 5-11-272, except the type I pail may be of the alternate design specified in 3.4. Nonmetallic sealing compound shall not be used in the construction of steel pails. The pails shall not leak when tested as specified in 4.4.4.

3.3.1 Side, bottom, and ears. The side, bottom, and ears shall be formed from one piece of material specified in 3.2.1.

3.3.2 Top edge. The top edge of the pail shall be beaded over the reinforcing wire specified in 3.2.2 to form a tightly rolled bead.

3.3.3 Bails. The bails shall be fabricated from wire specified in 3.2.2 and shall be similar to those shown on Drawing 5-11-272.

3.4 Type I, alternate design, aluminum. The type I, alternate design pail shall have a minimum capacity of 4 gallons. The pail body shall be formed from one piece of aluminum alloy sheet specified in 3.2.3 and shall have tapered sides. The bail shall be as specified in 3.3.3. The ears shall be aluminum and shall be of the offset type so designed to permit the bail to swing freely from one side of the pail to the other. The ears shall be riveted to the outside of the pail at points diametrically opposite each other. Each ear shall be attached by a minimum of two aluminum rivets.

3.4.1 Graduation marks. The aluminum pail shall have graduated markings indicating one gallon increments. The graduations shall be marked on the inside surface and shall extend around the circumference of the pail. The markings shall be free of any sharp corners and shall be clear and legible from a distance of 3 feet. The volume shall not vary more than $\pm$ 5 percent at any graduation when tested as specified in 4.4.4.

3.5 Marking for identification. Identification markings shall be on the outside bottom of the pails. Markings shall include the manufacturer's name and trademark and otherwise shall conform to the manufacturer's standard identification marking.

3.6 Finish.

3.6.1 Type I and II, steel pails. The completely fabricated steel pails, without bail attached, shall be cleaned and free from loose scale, rust, and other foreign matter. All surfaces of the pails, inside and outside, and the bail shall be completely coated with tin in accordance with type I or II of MIL-T-10727. The thickness of the coating shall be a minimum 0.0003 inches when tested as specified in 4.4.4. The coating shall be free from lumps, blisters, gritty areas, uncoated spots, acid, black spots, dross, and flux.

Source: https://assist.dla.mil -- Downloaded: 202640-16T14:17Z
Check the source to verify that this is the currentversion before use.

3.6.2 Aluminum pail. The type I aluminum pail shall have all rough and sharp edges, dirt, grease, or foreign matter removed in accordance with the manufacturer's standard practice. The pail body and ears shall be finished with a smooth commercial, standard bright finish and shall be uniform in appearance. The bail shall be cleaned and tin coated as specified in 3.6.1.

3.7 Workmanship. The pail shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in table I and dimensions specified in 4.4.3 and tested for the characteristics specified in 4.4.4. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:17Z
Check the source to verify that this is the current version before use.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of pails of one type. The sample unit shall be one pail. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish (general)	Finish not as specified	101	
	Not finished where required	102	
Steel pails	Flaking, peeling, blistered, rust, scale, areas of no coating, lumps, acid, black spots, dross, or flux	103	
	Tear drops or dripping on edges	104	
	Dirt, grit, or other foreign matter embedded in finish	105	
	Scratches through coating	106	
Aluminum pail	Not a smooth bright finish	107	
Design for steel pails	Nonmetallic sealing compound used	108	
Construction and workmanship	Any component dented, creased, or malformed	109	
	Any burrs, slivers, splinters, or sharp edges	110	
Body, steel pails (as applicable)	Not joined by lock seam	111	
	Top edge of pail not beaded over reinforcing wire	112	
Aluminum pails	Not one piece construction	113	
	Not tapered sides	114	

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:17Z
Check the source to verify that this is the current version before use.

TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Construction and workmanship (cont'd)			
Bottom, steel pails	Not lock seamed and crimped to sides	115	
Ears	Do not permit the loops of bail to swing freely	116	
	Not formed as specified	117	
	Not attached as specified	118	
	Missing	119	
Graduation marks for type I	Missing	120	
Bail	Missing	121	
Riveting	Rivet missing	122	
	Rivet loose	123	
Marking for identification	Omitted, missing, incomplete, or not applied in location specified		201

* 4.4.3 End item dimensional examination. The end items shall be examined for conformance to the dimension annotated with an asterisk on Drawing 5-11-272. Any annotated dimensions not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of pails of one type. The sample unit shall be one pail. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 End item testing. The end items shall be tested as specified in 4.5.1, 4.5.2, and 4.5.3, when applicable. The lot size shall be expressed in units of pails of one type. The sample unit shall be one pail. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:17Z
Check the source to verify that this is the current version before use.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.5 Methods of inspection.

* 4.5.1 Leakage. Pails shall be filled with water and allowed to stand for 5 minutes. The pails shall then be examined to determine compliance with 3.3.

4.5.2 Tin coating. Tin coating thickness shall be determined by the test procedure specified in MIL-T-10727 to determine compliance with 3.6.1.

4.5.3 Capacity test. The capacity of the type I aluminum pail shall be tested by measuring the volume of water at each graduated marking on the pail. The volumes shall be checked to determine compliance with 3.4.1.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.1.1 Level A packing. Six pails of one type shall be packed three in length, two in width, and one in depth in a snug-fitting crate conforming to type I, style A of MIL-C-52950. Alternatively, shipping containers may be used that conform to overseas type of PPP-B-601 or class 2, style 2 or 4 of PPP-B-621. Closure and strapping shall be in accordance with the appendix of the applicable container document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:17Z
Check the source to verify that this is the current version before use.

5.1.2 Level B packing.

* 5.1.2.1 Type I and II steel pails. Six pails of one type shall be packed three in length, two in width, and one in depth in a snug-fitting crate conforming to type I, style B of MIL-C-52950. Alternatively, shipping containers may be used that conform to domestic type, style A or B of PPP-B-601; class 1, style 2 or 4 of PPP-B-621; or class 1, style E of PPP-B-640.

5.1.2.2 Type I aluminum pail. Six pails shall be nested in the most compact manner and packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. The pails shall be positioned with the bail ears in the corners of the container. Fiberboard inserts shall be used to fill voids in each corner of the container. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636.

* 5.1.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2) the shipping container shall be V3c, V3s, or V4s fiberboard box fabricated and closed in accordance with PPP-B-636.

5.1.3 Commercial. Pails shall be packed in accordance with ASTM D 3951.

5.2 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The pails are intended for measuring flour and water for making bread. The pails are components of the Baking Plant, Field Portable, MIL-B-43322.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type of pail required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Selection of applicable level of packing (see 5.1).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.1.2.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:17Z
Check the source to verify that this is the current version before use.

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.4 Subject term (key word) listing.

Bread making
Pail, bakery
Pail, measuring

6.5 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD
Air Force - 84

User activity:

Navy - MC

Preparing activity:

Army - GL

Project No. 7240-0244

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:17Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

US Army Natick Research, Development, and
Engineering Center
Natick, MA 01760-5014

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
STRNC-ES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research, Development,
and Engineering Center
ATTN: STRNC-ES
Box 14A
Natick, MA 01760-5014

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:17Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-2067E	Pails, Metal, Measuring (For Bakeries)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil - Downloaded: 202610-16T14:17Z
Check the source to verify that this is the current version before use.